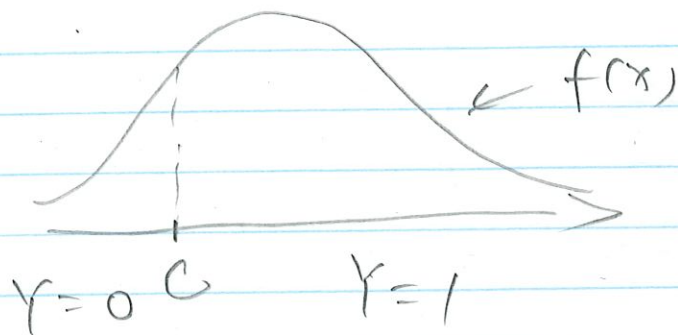


Truncated distribution

suppose $X \sim f(x)$ (e.g. normal, poisson)

let $Y = I(X > c)$, an indicator of $X > c$



Joint distribution of (X, Y) :

$$f_{xy}(x, y) = f(x) \cdot f(y|x)$$

where the conditional distribution $Y|X$ is

$$f(y|x) = \begin{cases} 1, & \text{if } y=1, x > c \\ 0, & \text{if } y=0, x \leq c \\ 0, & \text{if } y=1, x \leq c \\ 1, & \text{if } y=0, x > c \end{cases}$$

$$= [I(X > c)]^y [1 - I(X > c)]^{1-y}$$

Note: $0^0 = 1$

Marginal distribution of Y :

$$f_Y(y) = \int_{-\infty}^{+\infty} [I(X > c)]^y [1 - I(X > c)]^{1-y} f(x) dx$$

$$f_Y(1) = \int_{-\infty}^{+\infty} I(x > c) f(x) dx$$

$$= \int_c^{+\infty} f(x) dx = P(X > c)$$

$$f_Y(0) = \int_{-\infty}^c f(x) dx = P(X \leq c)$$

$$Y \sim \text{Bern}(P(X > c))$$

Conditional distribution of $X | Y$:

$$f_{X|Y}(x | Y=1) = \frac{f_{XY}(x, Y=1)}{f_Y(1)}$$

$$= \frac{f(x) \cdot I(x > c)}{P(X > c)}$$

$$= \begin{cases} \frac{f(x)}{\int_c^{+\infty} f(x) dx}, & \text{if } x > c \\ 0, & \text{o.w.} \end{cases}$$

Similarly,

$$f_{X|Y}(x | Y=0) = \frac{f(x)}{\int_{-\infty}^c f(x) dx} I(x \leq c)$$

A picture for truncated distribution

